ASSCA-Net: An Adaptive Spectral–Spatial Cooperative Attention Network for Hyperspectral Image Classification

Shufang Xu, *Member, IEEE*, Weiwen Xu, Shuyu Fei, Runhua Sheng, Lifeng Yu, and Hongmin Gao, *Senior Member, IEEE*

***Abstract*—Hyperspectral images (HSIs) possess fine spectral resolution. They can capture continuous and detailed spectral curves of ground objects, providing rich information for accurate classification. However, real-world scenes commonly suffer from diverse ground object morphology, spectral variability, and insufficient spatial–spectral feature interaction. These problems make effective joint feature extraction challenging. To address these issues, this article proposes an adaptive spatial–spectral collaborative attention network (ASSCA-Net). First, the network employs a spectral-guided spatial aggregation module (SSAM) at the input stage to suppress noise and enhance intraclass consistency. Then, a parallel dual-branch architecture is adopted. The spectral branch achieves differentiated band perception through an anchor-driven spectral attention mechanism (ASAM). The spatial branch captures long-range dependencies of heterogeneous targets via dynamic multiscale strip pooling (DMSP). Finally, a bidirectional collaborative attention fusion (BCAF) module enables deep interaction and adaptive fusion of spatial–spectral features. Experimental results on multiple public datasets show that the proposed method outperforms existing mainstream methods in accuracy.**

***Index Terms*—Adaptive network, attention mechanism, feature fusion, hyperspectral image (HSI) classification, spectral–spatial cooperation.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is a core task in remote sensing image interpretation. It has drawn widespread attention for a long time. Thanks to imaging spectrometers, which can acquire hundreds of continuous and narrow spectral bands, HSI can provide a near-continuous spectral curve for each pixel. This capability far exceeds that of multispectral images [1], [2]. This unique advantage gives HSI great application potential in many fields, such as geological exploration [3], [4], [5], agricultural monitoring [6], [7], [8], environmental monitoring [9], [10], [11], and military target detection [12], [13].

However, despite these significant advantages, HSI classification still faces many fundamental challenges. These include the Hughes phenomenon [14], [15], spectral variability [16], [17], and spatial scale diversity [18], [19]. Therefore, two core issues urgently need to be addressed in this field. One is designing spatial–spectral joint feature extraction models suitable for high-accuracy classification [20], [21]. The other is achieving deep collaboration between spectral and spatial information in complex scenes [22].

Early studies mainly relied on traditional machine learning methods such as support vector machines (SVMs) [23] and k-nearest neighbors (KNNs) [24]. They performed classification using handcrafted features, for example, Gabor features [25] and morphological profiles [26]. However, such methods struggle to capture the high-dimensional nonlinear structure of hyperspectral data. Moreover, utilizing spatial information often depends on complex postprocessing like Markov random fields [27]. As a result, their generalization ability is limited.

To break through these bottlenecks, deep learning techniques were introduced into HSI classification. Among them, convolutional neural networks (CNNs) became mainstream. From 1D-CNN [28] and 2D-CNN [29] to 3D-CNN [30] and the hybrid 2-D–3-D architecture HybridSN [31], CNNs gradually achieved automatic extraction of spectral, spatial, and joint spatial–spectral features. Subsequently, grouped separable convolution [32], multiscale depthwise separable convolution [33], and attention mechanisms [34], [35] were integrated to improve efficiency and feature selectivity. Although CNNs perform well in local feature extraction, their inherent local receptive field limits their ability to capture global spectral dependencies and long-range spatial context. In addition, their modeling capability for spectral variability remains limited.

To compensate for the global modeling deficiency of CNNs, Vision Transformers (ViTs) have drawn attention due to their long-range dependency capability. SpectralFormer [36] first applied Transformers to HSI classification. It learns spectral sequence representations through grouped spectral embedding. However, pure Transformers lack spatial local inductive bias and suffer from high computational complexity. Later improvements, such as SSFTT [37], GAHT [38],

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Shufang Xu, Weiwen Xu, Shuyu Fei, and Hongmin Gao are with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: xushufang@hhu.edu.cn; 241607010006@hhu.edu.cn; feishuyu@hhu.edu.cn; gaohongmin@hhu.edu.cn).

Runhua Sheng is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: shengrunhua@whu.edu.cn).

Lifeng Yu is with the School of Information and Design, Zhejiang Industry Polytechnic College, Shaoxing 312000, China (e-mail: 20040015@zjipc.edu.cn).

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and Swin Transformer [39], have alleviated these problems to some extent. Yet, they still struggle to fully learn joint spatial–spectral representations. Multiscale low-rank sparse attention [40] and channel-grouped gated self-attention [41] have strengthened spectral difference perception. However, they still have clear deficiencies in spatial adaptive modeling.

To break through this bottleneck, CNN–Transformer hybrid architectures emerged. Basic dual-branch architectures, including DBCTNet [42], MorphFormer [43], GSC-ViT [44], DBSSFNet [45], and S^{2}DBFT [46], enhance spatial–spectral feature interaction from different perspectives. CAEVT [47] and PSFormer [48] combine autoencoders and superpixel segmentation, respectively. MsGsFNet [49] focuses on fusing multiscale spatial and global spectral information. Dynamic direction-aware convolution [50] and multiscale strip pooling [46] explore spatial adaptation. The optimal transport framework [51], multiscale dual aggregation network [52], and PID-inspired multiscale attention [53] attempt fine-grained spectral modeling and feature extraction. Building on these, STM-Net [54], HyperTransXNet [55], and MSDCA [56] further strengthen dual-branch collaboration and cross-attention fusion. Meanwhile, lightweight memory-driven self-attention [57], dual-attention convolution simplified Transformers [58], manifold geometry-based self-supervised learning [59], and pooling attention fusion [60] strike a balance between efficiency and accuracy.

Although the above hybrid architectures have made significant progress, they still generally have limitations in the following three aspects. First, the spatial dependency modeling capability for atypically shaped objects, such as strip-like and winding ground objects, is insufficient. Existing convolution kernels are limited by fixed geometric shapes. They cannot adapt their receptive fields according to local structures. Although dynamic direction-aware convolution [50] and multiscale strip pooling [46] have been proposed, they mostly adopt a hard binary choice and lack a continuous mixing strategy. Second, the object-dependent nature of spectral band importance is ignored. Most existing methods cause all ground objects to share the same set of band weights. This makes it difficult to handle the "same object and different spectra" phenomenon. The optimal transport framework [51] and multiscale dual aggregation network [52] have provided new ideas. Yet, object-adaptive variation of band weights remains a challenging problem. Third, the depth of spatial–spectral feature fusion is insufficient. Existing dual-branch fusion networks [45], [46] and progressive multiscale multiattention fusion [53] mostly adopt concatenation or unidirectional attention. They fail to achieve true bidirectional collaborative enhancement.

To address these three challenges, this article proposes an adaptive spatial–spectral collaborative attention network (ASSCA-Net). The solutions are as follows.

1) To address the insufficient spatial dependency modeling capability for atypically shaped objects such as strip-like and winding ground objects, on the one hand, a spectral-guided spatial aggregation module (SSAM) is placed at the input stage. This module performs spectral-similarity-guided spatial pre-extraction, suppressing noise and enhancing intraclass spectral consistency. On the other hand, a dynamic multiscale strip pooling (DMSP) module is designed. It adaptively selects horizontal or vertical pooling directions based on local gradient orientation. This achieves precise modeling of long-range spatial dependencies for heterogeneous targets.
2) To address the ignored object-dependent nature of spectral band importance, an anchor-driven spectral attention mechanism (ASAM) module is designed. It introduces differentiable optimal transport to cluster spectra into learnable anchors. This enables each ground object to acquire differentiated band perception capability. It improves robustness to the "same object and different spectra" phenomenon.
3) To address the problem that spectral and spatial features are often processed independently and their fusion relies only on simple concatenation or weighted summation, lacking deep interaction, a bidirectional collaborative attention fusion (BCAF) module is designed. Through bidirectional cross-attention, the two types of features can query each other and enhance each other. Then, via adaptive weighted fusion, a more discriminative unified spatial–spectral representation is generated.

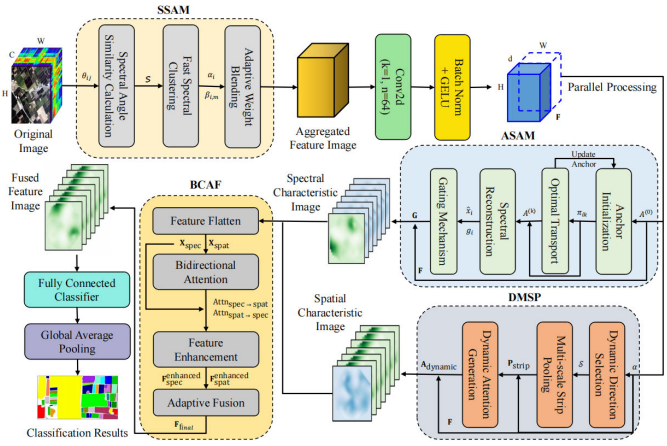


Fig. 1. Overall framework of the ASSCA-Net model.

II. Proposed Method

A. Overall Framework

The proposed ASSCA-Net takes an HSI patch $X \in \mathbb{R}^{C\times H\times W}$ as input in an end-to-end manner. It outputs a K-dimensional class probability vector. As shown in Fig. 1, the network follows three sequentially connected core processing stages: spatial aggregation, spatial–spectral dual-branch feature extraction, and collaborative attention fusion.

In the spatial aggregation stage, the input X is first processed by the SSAM. This yields a spatially aggregated map $X' \in \mathbb{R}^{C\times H\times W}$. Subsequently, X' passes through a 1×1 convolution that compresses the channel number to $d = 64$. This generates a compact feature map $F \in \mathbb{R}^{d\times H\times W}$.

Then F is fed simultaneously into two parallel branches. The spectral branch adopts ASAM. It outputs $F_{\text{spectral}} \in \mathbb{R}^{d\times H\times W}$. The spatial branch adopts DMSP. It outputs $F_{\text{spatial}} \in \mathbb{R}^{d\times H\times W}$. It should be noted that SSAM serves as an

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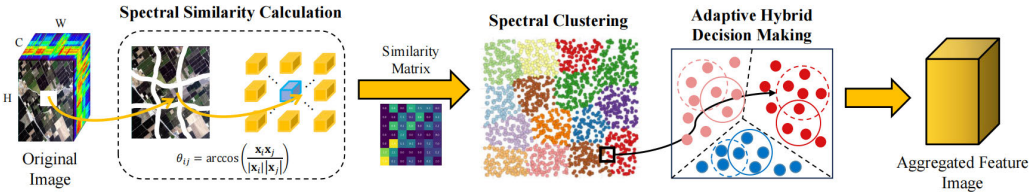


Fig. 2. Schematic of the SSAM.

input-level isotropic preprocessing module for spectral purity enhancement, while DMSP is a feature-level anisotropic module for directional spatial modeling. The two are complementary rather than redundant.

Finally, F_{spectral} and F_{spatial} are fed into the BCAF. This produces a unified spatial–spectral feature $F_{\text{final}} \in \mathbb{R}^{d\times H\times W}$. After global average pooling and a fully connected layer, the final classification probability vector is output.

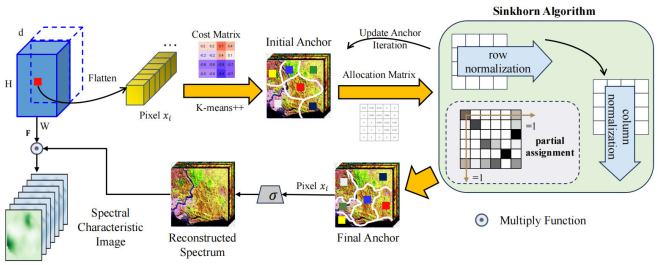


Fig. 3. Schematic of the ASAM module.

B. Spectral-Guided Spatial Aggregation Module

The SSAM module is positioned at the front of the network. It aims to enhance the spectral consistency of same-class ground objects and suppress noise through spatial aggregation guided by spectral similarity. As shown in Fig. 2, this module takes the original HSI patch $X \in \mathbb{R}^{C\times H\times W}$ as input. It outputs a spatially aggregated feature map $X' \in \mathbb{R}^{C\times H\times W}$. This provides the subsequent dual-branch feature extraction with an initial representation that has enhanced intraclass compactness.

For any two pixels x_i and x_j in the input image patch ($i, j \in \{1,\ldots,N, N = H\times W$), their spectral angular similarity (SAS) is computed as follows:

$$\text{SAS}\left(\mathbf{x}_i,\mathbf{x}_j\right) = \exp\left(-\frac{\theta_{ij}}{\sigma}\right)$$
$$\theta_{ij} = \arccos\left(\frac{\mathbf{x}_i\mathbf{x}_j}{\|\mathbf{x}_i\|\left\|\mathbf{x}_j\right\|}\right) \tag{1}$$

where σ is a temperature parameter that controls the decay rate of the SAS. Based on the computed similarity matrix $S \in \mathbb{R}^{N\times N}$, a fast spectral clustering algorithm is employed to divide all pixels into K spectral clusters $\{C_1,\ldots,C_K\}$. Pixels within the same cluster share similar spectra, while those in different clusters exhibit significant differences.

For a pixel x_i belonging to cluster C_k, M neighboring pixels $x_{j_1},\ldots,x_{j_M}$ are randomly selected from the same cluster. The mixing weights are adaptively assigned according to the spectral similarity

$$\beta_{i,m} = \frac{\exp\left(\gamma\cdot\text{SAS}\left(\mathbf{x}_i,\mathbf{x}_{j_m}\right)\right)}{\sum_{n=1}^{M}\exp\left(\gamma\cdot\text{SAS}\left(\mathbf{x}_i,\mathbf{x}_{j_n}\right)\right)}$$
$$\alpha_i = 1-\sum_{m=1}^{M}\beta_{i,m} \tag{2}$$

where α_i is the reserved weight for the original pixel to prevent over-smoothing and $\beta_{i,m}$ are the adaptive weights assigned to neighboring pixels based on spectral similarity. They satisfy $\alpha_i + \sum_{m=1}^{M}\beta_{i,m} = 1$, ensuring a convex combination. γ is a temperature parameter controlling the sharpness of the softmax distribution. The aggregated pixel is

$$\mathbf{x}_i' = \alpha_i\mathbf{x}_i + \sum_{m=1}^{M}\beta_{i,m}\mathbf{x}_{j_m}. \tag{3}$$

Notably, SSAM is isotropic and direction-agnostic, as its aggregation is driven solely by spectral similarity without considering spatial orientation. X' then enters the dual-branch feature extraction stage. It provides cleaner and more discriminative initial representations for the spectral branch ASAM and the spatial branch DMSP.

C. Anchor-Driven Spectral Attention Mechanism

The ASAM module aims to address the "same object and different spectra" problem in HSIs. It performs pointwise modeling and refinement of the spectral feature of each pixel to enhance spectral discriminability. As shown in Fig. 3, the input to this module is the feature map $F \in \mathbb{R}^{d\times H\times W}$ ($d = 64$). The output is a spectrally enhanced feature map $F_{\text{spectral}} \in \mathbb{R}^{d\times H\times W}$. Unlike global pooling, ASAM retains the full spatial dimension. This allows each pixel to receive calibration from the global spectral template while preserving spatial information.

1) Anchor Initialization: The feature map $F \in \mathbb{R}^{d\times H\times W}$ is flattened into $N = H\times W$ spectral vectors of dimension d, denoted as $\{x_1,\ldots,x_N\}$. K-means++ is adopted to initialize K anchors $A^{(0)} = \{a_1^{(0)},\ldots,a_K^{(0)}$, where each $a_k^{(0)} \in \mathbb{R}^d$. The greedy strategy of K-means++ provides stable initial cluster centers, avoiding the uncertainty of random initialization. Afterward, these K anchors are stored as global parameters and shared by all input patches. K is set to the number of dataset classes.

2) Optimal Transport Soft Assignment: A cost matrix $C \in \mathbb{R}^{N\times K}$ is constructed. Its element C_{ik} denotes the cost of assigning pixel i to anchor k. It integrates Euclidean distance and cosine distance

$$C_ik = \|x_i - a_k\|_2^2 + \beta\cdot(1-\cos(x_i,a_k)) \tag{4}$$

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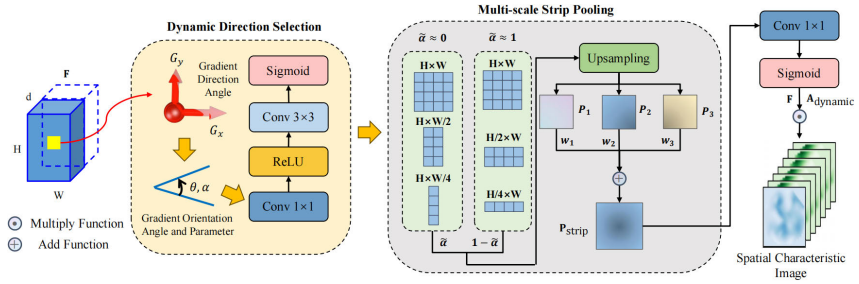


Fig. 4. Schematic of the DMSP module.

where $\|\cdot\|_2$ denotes the L2 norm, $\cos(\cdot,\cdot)$ denotes cosine similarity, and β is a hyperparameter that balances the two terms. The Euclidean distance captures magnitude differences, while the cosine distance captures spectral shape differences.

Based on the cost matrix, the entropy-regularized optimal transport problem is solved to obtain a soft assignment matrix $\pi \in \mathbb{R}^{N\times K}$

$$\min_{\pi\in R^{(N\times K)}} \langle \pi, \mathrm{C}\rangle_F + \varepsilon \cdot H(\pi)$$
$$\text{subject to}: \ \pi \cdot \mathbf{1}_K = \frac{1}{N}\mathbf{1}_N, \quad \pi^\top \cdot \mathbf{1}_N = \frac{1}{K}\mathbf{1}_K, \ \pi \geq 0 \quad (5)$$

where $\langle\cdot,\cdot\rangle_F$ is the Frobenius inner product, $H(\pi) = -\sum_{i,k}\pi_{ik}(\log\pi_{ik}-1)$ is the entropy of the transport plan, and $\varepsilon > 0$ is the regularization coefficient.

The solution to this problem can be efficiently obtained via the Sinkhorn–Knopp iterative algorithm, which alternately normalizes rows and columns. The row constraint $\pi\cdot\mathbf{1}_K = (1/N)\mathbf{1}_N$ means each pixel i must distribute its total probability mass of $1/N$ to all anchors. The column constraint $\pi^\top \cdot \mathbf{1}_N = (1/K)\mathbf{1}_K$ requires each anchor k to receive a total probability mass of $1/K$ from all pixels.

3) Anchor Update: After obtaining the optimal assignment matrix π for the current patch, the anchors are locally updated according to the soft assignments. They are moved toward the data center of the pixels within the current patch

$$a_k = \frac{\left(\Sigma_{i=1}^N \pi_{ik} * x_i\right)}{\left(\Sigma_{i=1}^N \pi_{ik}\right)}. \quad (6)$$

Equation (6) computes the weighted average of all pixel spectral vectors, using the belonging probability π_{ik} as the weight. The updated anchors temporarily replace the global anchors within the current patch.

4) Spectral Reconstruction and Gating: For each pixel x_i in the current patch, its reconstructed spectrum is obtained by a weighted sum over the K updated anchors using its belonging probabilities

$$\hat{x}_i = \Sigma_{k=1}^K \pi_{ik} * a_k. \quad (7)$$

To balance the original and reconstructed features, the residual between the original and reconstructed spectra is used to generate a gating signal. This signal adaptively modulates the strength of the original features

$$g_i = \sigma\left(\frac{(\hat{x}_i - x_i)}{\tau}\right) \quad (8)$$

where σ is the Sigmoid function and τ is a temperature parameter. All gate values $g_i \in (0,1)$ are rearranged according to their spatial positions to obtain a spatial gating map $G \in \mathbb{R}^{1\times H\times W}$

$$\mathbf{G} = \text{Reshape}\left(\{g_1, g_2, \ldots, g_N\}\right). \quad (9)$$

$\mathbf{G}$ is broadcast along the channel dimension to the same d channels as the input feature F. This yields the final gating map $\mathbf{g} \in \mathbb{R}^{d\times H\times W}$. It is then element-wise multiplied with the original feature to produce the spectrally enhanced feature $F_{\text{spectral}} \in \mathbb{R}^{d\times H\times W}$

$$\mathbf{F}_{\text{spectral}} = \mathbf{F} \odot \text{Broadcast}(\mathbf{G}, d) \quad (10)$$

where $\odot$ denotes the elementwise multiplication. Through this operation, typical sample features consistent with the global spectral template are preserved or enhanced. Outlier or noisy sample features that deviate from the global template are suppressed.

D. Dynamic Multiscale Strip Pooling

The DMSP module aims to effectively capture the long-range spatial dependencies of strip-like and winding ground objects in HSIs. Different from the isotropic SSAM, DMSP is explicitly anisotropic and direction-aware, serving as a complementary module to model long-range contextual dependencies along adaptively selected orientations.

As shown in Fig. 4, the input to this module is the feature map $F \in \mathbb{R}^{d\times H\times W}$ ($d = 64$). The output is a spatially enhanced feature map $F_{\text{spatial}} \in \mathbb{R}^{d\times H\times W}$. By dynamically selecting the pooling direction based on the local gradient orientation and combining a multiscale strategy with adaptive convolution, DMSP achieves accurate modeling of anisotropic spatial structures.

1) Dynamic Direction Selection: The Sobel operator is used to compute the horizontal gradient G_x and the vertical gradient G_y. The gradient orientation angle at each position is obtained as $\theta(h,w) = \arctan2(G_y(h,w), G_x(h,w))$. To avoid hard direction switching, a continuous direction parameter is defined

$$\alpha(h,w) = \frac{1+\cos(2\theta(h,w))}{2} \in [0,1]. \quad (11)$$

When $\alpha \approx 1$, horizontal strip pooling is preferred. When $\alpha \approx 0$, vertical strip pooling is preferred. To enhance robustness, the α map is refined through two convolutional layers

$$\tilde{\alpha} = \sigma\left(\text{Conv}_{k3}\left(\text{ReLU}\left(\text{Conv}_{k1}(\alpha)\right)\right)\right). \quad (12)$$

Here, Conv_{k1} and Conv_{k3} denote the convolutional layers with kernel sizes of 1×1 and 3×3, respectively. This allows the

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network to automatically learn how to adjust direction weights according to the local context.

The continuous mixing weight α enables DMSP to handle arbitrary orientations, as any diagonal direction can be represented by a weighted combination of horizontal and vertical pooling, avoiding the hard switching boundaries inherent in discrete direction selection.

2) Multiscale Strip Pooling: Let the set of scales be $S = \{s_1, s_2, s_3\}$. For horizontal pooling, $s_1 = H$, $s_2 = H/2$, and $s_3 = H/4$. For vertical pooling, $s_1 = W$, $s_2 = W/2$, and $s_3 = W/4$. For each scale s_i, the spatial size of the feature map is first scaled to s_i via average pooling. Then horizontal strip pooling and vertical strip pooling are performed, respectively,

$$P_H^{(s)}(c,1,w) = \frac{1}{s}\sum_{h=1}^{s}\mathbf{F}^{(s)}(c,h,w)$$
$$P_V^{(s)}(c,h,1) = \frac{1}{s}\sum_{w=1}^{s}\mathbf{F}^{(s)}(c,h,w) \tag{13}$$

here, $F^{(s)}$ denotes the feature map adjusted to scale s. $P_H^{(s)}$ compresses the s pixels along the height direction into a single value. $P_V^{(s)}$ compresses the s pixels along the width direction into a single value.

Subsequently, the horizontal and vertical pooling results from the three scales are fused through a weighted combination according to the direction weight $\tilde{\alpha}$

$$\mathbf{P}_{\text{strip}} = \sum_{i=1}^{3} w_i \cdot \begin{bmatrix} \tilde{\alpha}\cdot \text{Upsample}\left(P_H^{(s_i)}\right) + \\ (1-\tilde{\alpha})\cdot \text{Upsample}\left(P_V^{(s_i)}\right) \end{bmatrix}. \tag{14}$$

Here, w_i is the learnable scale weights. Upsample$(\cdot)$ upsamples the pooling result at each scale back to the original spatial size $H \times W$. $\tilde{\alpha}$ controls the mixing ratio between the horizontal and vertical directions.

The three scales are designed to capture complementary information: the largest scale provides global context, the medium scale captures regional structures, and the smallest scale preserves fine-grained local details. The learnable weights w_i automatically adjust the contribution of each scale according to the input characteristics, effectively mitigating potential redundancy.

3) Dynamic Spatial Attention Generation: The multiscale strip pooling output P_{strip} is transformed into a spatial attention map through a 1×1 convolution and Sigmoid function. It is then elementwise multiplied with the input feature to produce the spatially enhanced feature

$$\mathbf{F}_{\text{spatial}} = \mathbf{F} \odot \sigma\left(\text{Conv}_{1\times1}\left(\mathbf{P}_{\text{strip}}\right)\right). \tag{15}$$

The final output $F_{\text{spatial}} \in \mathbb{R}^{d\times H\times W}$ maintains exactly the same spatial size as the input. It then enters the BCAF module for collaborative fusion with the output of the spectral branch.

E. Bidirectional Collaborative Attention Fusion

The BCAF module aims to achieve deep interaction and adaptive fusion of spectral and spatial features. As shown in Fig. 5, this module takes the spectral branch output $F_{\text{spectral}} \in \mathbb{R}^{d\times H\times W}$ and the spatial branch output $F_{\text{spatial}} \in \mathbb{R}^{d\times H\times W}$ as input. It outputs a unified spatial–spectral feature $F_{\text{final}} \in \mathbb{R}^{d\times H\times W}$. Through a bidirectional cross-attention mechanism, the spectral features query spatial structural information, and the spatial features query spectral discriminative information. After mutually enhancing each other, they undergo adaptive weighted fusion. This generates a more discriminative joint representation.

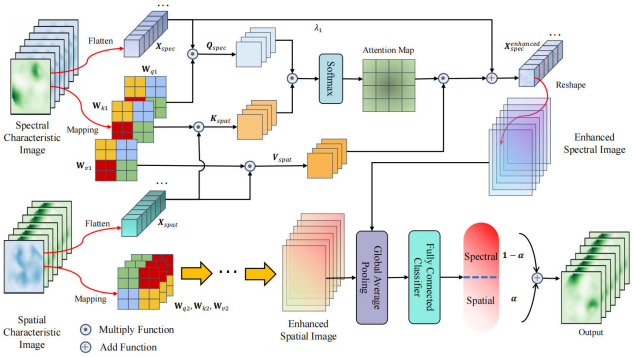


Fig. 5. Schematic of the BCAF module.

First, the input spectrally enhanced feature F_{spectral} and spatially enhanced feature F_{spatial} are flattened in the spatial dimension. They are converted into two feature sequences

$$\mathbf{X}_{\text{spec}} = \text{Flatten}\left(\mathbf{F}_{\text{spectral}}\right) \in \mathbb{R}^{N\times d}$$
$$\mathbf{X}_{\text{spat}} = \text{Flatten}\left(\mathbf{F}_{\text{spatial}}\right) \in \mathbb{R}^{N\times d}. \tag{16}$$

Next, for each direction, we define learnable projection matrices. For the spectral-querying–spatial direction, we use $W_{q1}, W_{k1}, W_{v1} \in \mathbb{R}^{d\times d_k}$. For the spatial-querying-spectral direction, we use $W_{q2}, W_{k2}, W_{v2} \in \mathbb{R}^{d\times d_k}$. Here, d_k is the dimension of the attention head.

In the spectral-querying–spatial direction, the projection matrices are first used to generate the query, key, and value vectors

$$\mathbf{Q}_{\text{spec}} = \mathbf{X}_{\text{spec}}\mathbf{W}_{q1}$$
$$\mathbf{K}_{\text{spat}} = \mathbf{X}_{\text{spat}}\mathbf{W}_{k1}$$
$$\mathbf{V}_{\text{spat}} = \mathbf{X}_{\text{spat}}\mathbf{W}_{v1}. \tag{17}$$

Then, the attention weight matrix is computed and used to take a weighted sum of the value vectors. This yields the contextual information that the spectral features extract from the spatial features

$$\text{Attn}_{\text{spec}\to\text{spat}} = \text{Softmax}\left(\frac{\mathbf{Q}_{\text{spec}}\mathbf{K}_{\text{spat}}^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{V}_{\text{spat}} \tag{18}$$

where d_k denotes the dimension of the query and key projections in the attention head.

Symmetrically, in the spatial-querying-spectral direction, the goal is for each spatial feature position to attend to all spectral feature positions. This allows it to acquire discriminative spectral information.

After obtaining the attention outputs from both directions, we add them back to the original feature sequences via residual connections. Learnable gating parameters λ_1 and λ_2 control the enhancement strength

$$\mathbf{X}_{\text{spec}}^{\text{enhanced}} = \mathbf{X}_{\text{spec}} + \lambda_1 \cdot \text{Attn}_{\text{spec}\to\text{spat}}$$
$$\mathbf{X}_{\text{spat}}^{\text{enhanced}} = \mathbf{X}_{\text{spat}} + \lambda_2 \cdot \text{Attn}_{\text{spat}\to\text{spec}}. \tag{19}$$

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Here, λ_1 and λ_2 are the learnable gating parameters. This design allows the model to adaptively decide how much information to absorb from the other feature branch.

After $X_{\text{spec}}^{\text{enhanced}}$ and $X_{\text{spat}}^{\text{enhanced}}$ are reshaped back to the spatial dimensions to obtain $F_{\text{spec}}^{\text{enhanced}}$ and $F_{\text{spat}}^{\text{enhanced}}$, and global average pooling is applied to these two enhanced feature maps. This compresses the spatial dimension

$$\begin{aligned}\mathbf{z}_{\text{spec}} &= \text{GAP}\left(\mathbf{F}_{\text{spec}}^{\text{enhanced}}\right) \in \mathbb{R}^d \\ \mathbf{z}_{\text{spat}} &= \text{GAP}\left(\mathbf{F}_{\text{spat}}^{\text{enhanced}}\right) \in \mathbb{R}^d.\end{aligned} \tag{20}$$

The two pooled vectors are concatenated. A lightweight fully connected layer then learns the fusion weight

$$\alpha = \sigma\left(\mathbf{W}_f \cdot \left[\mathbf{z}_{\text{spec}}; \mathbf{z}_{\text{spat}}\right] + b_f\right) \tag{21}$$

where $W_f \in \mathbb{R}^{1\times 2d}$, $b_f \in \mathrm{R}$, σ is the Sigmoid function, and $\alpha \in [0, 1]$. The fused feature is obtained by weighted summation

$$\mathbf{F}_{\text{final}} = \alpha \cdot \mathbf{F}_{\text{spec}}^{\text{enhanced}} + (1-\alpha) \cdot \mathbf{F}_{\text{spat}}^{\text{enhanced}}. \tag{22}$$

Finally, $F_{\text{final}} \in \mathbb{R}^{d\times H\times W}$ goes through global average pooling and a fully connected layer. The output is a classification probability vector of K classes.

F. Joint Loss Function

To guide the network to learn highly discriminative feature representations and improve its generalization ability under limited labeled samples, a joint loss function is designed. It consists of two terms: the standard multiclass cross-entropy loss L_{CE} and an intraclass spectral consistency regularization loss L_{reg}. The cross-entropy loss is defined as follows:

$$\mathrm{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k}\log\left(\hat{y}_{i,k}\right) \tag{23}$$

where N is the number of samples in a batch, K is the number of classes, $y_{i,c}$ is the ground-truth label indicator, and $\hat{y}_{i,c}$ is the predicted probability for class c.

The regularization loss L_{reg} acts on the output features of the ASAM module. It aims to encourage pixels of the same class to be more compactly distributed in the feature space. Specifically, for N samples in a batch, the mean squared Euclidean distance is computed between the features extracted and pooled by ASAM and the spectral anchor vectors of their corresponding classes

$$L_{\text{reg}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{z}_i - \mathbf{a}_{C_i}\right\|_2^2. \tag{24}$$

Here, z_i is the feature representation of sample i after processing by the ASAM module. a_{C_i} is the spectral anchor vector in the ASAM module that is most strongly associated with the true class C_i of sample i.

The final joint loss function is the weighted sum of the above two losses

$$L = L_{\text{CE}} + \lambda \cdot L_{\text{reg}} \tag{25}$$

where λ is a hyperparameter that balances classification accuracy and feature compactness.

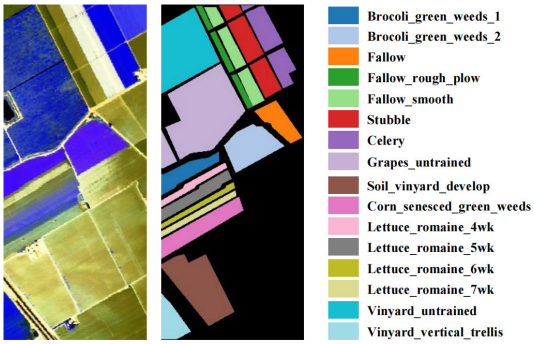


Fig. 6. (a) False-color composite of the Salinas scene. (b) Ground-truth map containing 16 mutually exclusive land-cover categories.

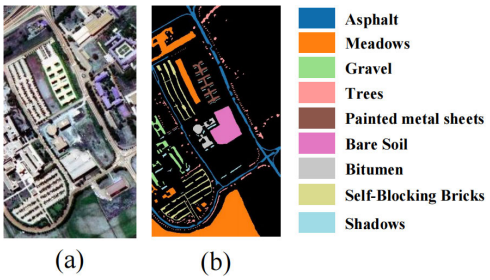


Fig. 7. (a) False-color composite of the PaviaU scene. (b) Ground-truth map containing nine mutually exclusive land-cover categories.

By optimizing this joint objective end-to-end, ASSCA-Net can not only learn accurate classification decision boundaries, but also form highly discriminative and robust spatial–spectral joint features. This enables stable and superior classification performance in complex scenes.

III. Experimental Results

This section comprehensively evaluates ASSCA-Net on four public hyperspectral datasets. It covers dataset introduction, experimental settings, parameter sensitivity analysis, ablation study, comparative experiments, and performance and visualization analysis.

A. Datasets

1) *Salinas Dataset:* This dataset was acquired by the AVIRIS imaging spectrometer over the Salinas Valley, CA, USA. The spatial resolution is 3.7 m. After removing water absorption bands, 204 bands remain. The image size is 512 × 217 pixels. It contains 16 land-cover classes, with a total of 54 129 labeled pixels, see Table I for details. Fig. 6(a) and (b) shows the false-color composites of the image and the corresponding ground-truth map.
2) *PaviaU Dataset:* This dataset was acquired by the ROSIS airborne imaging spectrometer over the University of Pavia, Italy. The spatial resolution is 1.3 m. After removing noisy bands, 103 bands remain. The image size is 610 × 340 pixels. It contains nine land-cover classes, with a total of 42 776 labeled pixels, see Table II for details. Fig. 7(a) and (b) shows the false-color composites of the image and the corresponding ground-truth map.

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TABLE I

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES PER LAND-COVER CLASS IN THE SALINAS DATASET

Class No.	Class Name	Training Samples	Validation Samples	Test Samples
C1	Brocoli_ green_weeds_1	20	20	1969
C2	Brocoli_ green_weeds_2	37	37	3652
C3	Fallow	20	20	1936
C4	Fallow_ rough_plow	14	14	1366
C5	Fallow_smooth	27	27	2624
C6	Stubble	40	39	3880
C7	Celery	36	36	3507
C8	Grapes_ untrained	112	113	11046
C9	Soil_vinyard_ develop	62	62	6079
C10	Corn_senesced_ green_weeds	33	33	3212
C11	Lettuce_ romaine_4wk	11	10	1047
C12	Lettuce_ romaine_5wk	20	19	1888
C13	Lettuce_ romaine_6wk	9	9	898
C14	Lettuce_ romaine_7wk	10	11	1049
C15	Vinyard_ untrained	72	73	7123
C16	Vinyard_ vertical_trellis	18	18	1771
	Total	541	541	53047

TABLE II

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES PER LAND-COVER CLASS IN THE PAVIAU DATASET

Class No.	Class Name	Training Samples	Validation Samples	Test Samples
C1	Asphalt	66	66	6499
C2	Meadows	186	187	18276
C3	Gravel	21	21	2057
C4	Trees	30	31	3003
C5	Painted metal sheets	14	13	1318
C6	Bare Soil	50	50	4929
C7	Bitumen	13	14	1303
C8	Self-Blocking Bricks	37	37	3608
C9	Shadows	10	9	928
	Total	427	428	41921

3) *WHU-Hi-HongHu Dataset:* Acquired in 2017 in Honghu, Hubei, China, using a DJI Matrice 600 Pro UAV equipped with a Headwall nano-hyperspectral sensor. The wavelength range is 400–1000 nm, with 270 bands. The image size is 940 × 475 pixels, and the spatial resolution is about 0.043 m. It contains 22 complex agricultural classes, see Table III for details. Fig. 8(a) and (b) shows the false-color composites of the image and the corresponding ground-truth map.
4) *WHU-Hi-LongKou Dataset:* Acquired in 2018 in Longkou, Hubei, China. It also used the DJI Matrice 600 Pro and Headwall sensor, with a wavelength range of 400–1000 nm and 270 bands. The image size is 550 × 400 pixels, and the spatial resolution is about 0.463 m. It contains nine land-cover classes, see Table IV for details. Fig. 9(a) and (b) shows the false-color composites of the image and the corresponding ground-truth map.

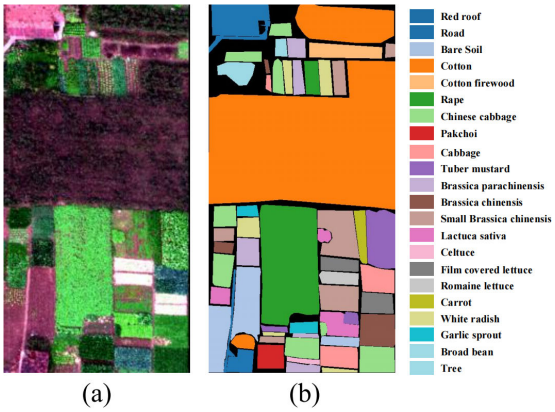


Fig. 8. (a) False-color composite of the WHU-Hi-HongHu scene. (b) Ground-truth map containin 22 mutually exclusive land-cover categories.

TABLE III

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES PER LAND-COVER CLASS IN THE WHU-HI-HONGHU DATASET

Class No.	Class Name	Training Samples	Validation Samples	Test Samples
C1	Red roof	140	141	13760
C2	Road	35	35	3442
C3	Bare soil	218	219	21384
C4	Cotton	1632	1633	160020
C5	Cotton firewood	62	62	6094
C6	Rape	445	446	43666
C7	Chinese cabbage	241	241	23631
C8	Pakchoi	41	40	3973
C9	Cabbage	108	109	10602
C10	Tuber mustard	124	124	12146
C11	Brassica parachinensis	110	110	10795
C12	Brassica chinensis	89	90	8775
C13	Small Brassica chinensis	225	225	22057
C14	Lactuca sativa	74	73	7209
C15	Celtuce	10	10	982
C16	Film covered lettuce	73	72	7117
C17	Romaine lettuce	30	30	2950
C18	Carrot	32	32	3153
C19	White radish	87	87	8538
C20	Garlic sprout	35	35	3416
C21	Broad bean	14	13	1301
C22	Tree	41	40	3959
	Total	3866	3867	378780

B. Experimental Settings

1) *Evaluation Metrics:* Overall accuracy (OA), average accuracy (AA), Kappa coefficient (κ), and accuracy of each class (AEC) are adopted as evaluation metrics. All results are based on ten independent random runs and

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TABLE IV
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES PER LAND-COVER CLASS IN THE WHU-HI-LONGKOU DATASET

Class No.	Class Name	Training Samples	Validation Samples	Test Samples
C1	Corn	34	35	34442
C2	Cotton	8	9	8357
C3	Sesame	3	3	3025
C4	Broad-leaf soybean	63	63	63086
C5	Narrow-leaf soybean	4	4	4143
C6	Rice	12	12	11830
C7	Water	67	67	66922
C8	Roads and houses	7	7	7110
C9	Mixed weed	6	5	5218
	Total	204	205	204133

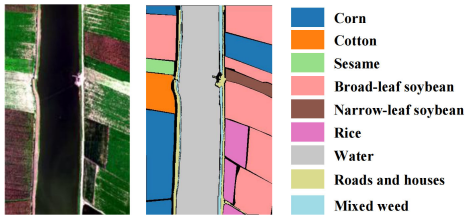


Fig. 9. (a) False-color composite of the WHU-Hi-LongKou scene. (b) Ground-truth map containing nine mutually exclusive land-cover categories.

reported as mean ± standard deviation. In addition, the classification maps are visually compared.

2) *Experimental Environment:* All experiments are conducted on a machine equipped with an Intel Core i9 14900HX CPU and an NVIDIA GeForce RTX 4060 GPU.
3) *Hyperparameter Settings:* The AdamW optimizer is used, with an initial learning rate of 0.001 and a weight decay of 0.05. The training batch size is 128, and the test batch size is 64. The models are trained for 200 epochs, and the learning rate is decayed by a factor of 0.5 at the 80th, 140th, and 180th epoch. All reported results are averaged over ten independent runs with different random seeds to ensure statistical reliability.
4) *Training/Validation Split Ratios:* For the Salinas, PaviaU, and WHU-Hi-HongHu datasets, 1% of samples per class are used for training and 1% for validation; for the WHU-Hi-LongKou dataset, 0.1% per class are used. Both the comparative experiments and the ablation study adopt the optimal patch size and λ for each dataset. All compared methods share the same data splits and evaluation protocol.

C. Comparative Experiments

To evaluate the effectiveness of the proposed method, ASSCA-Net is compared with various HSI classification methods based on CNNs and Transformers. The compared methods include HybridSN [31], SPRN [35], SpectralFormer [36], SSFTT [37], GAHT [38], MorphFormer [43], GSC-ViT [44], CAEVT [47], and MsGsFNet [49]. Four public datasets—Salinas, Pavia University, WHU-Hi-HongHu, and WHU-Hi-LongKou—are used. The classification accuracies are reported in Tables V–VIII, and the corresponding classification maps are shown in Figs. 10–13.

As shown in Fig. 10 and Table V, this agricultural scene contains 16 land-cover classes. ASSCA-Net ranks first in OA,

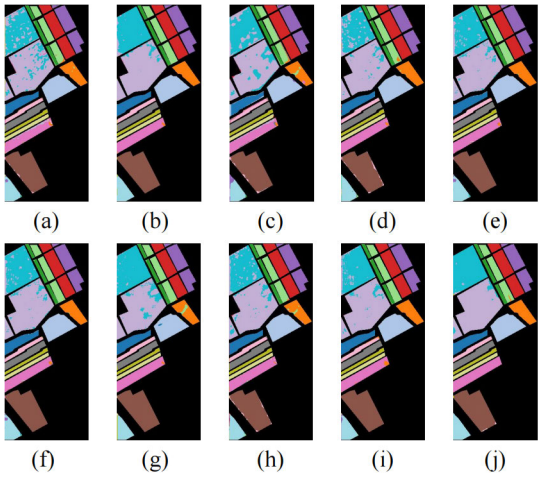


Fig. 10. Classification maps of different methods on the Salinas dataset. (a) HybridSN. (b) SPRN. (c) SpectralFormer. (d) SSFTT. (e) GAHT. (f) MorphFormer. (g) GSC-ViT. (h) CAEVT. (i) MsGsFNet. (j) ASSCA-Net.

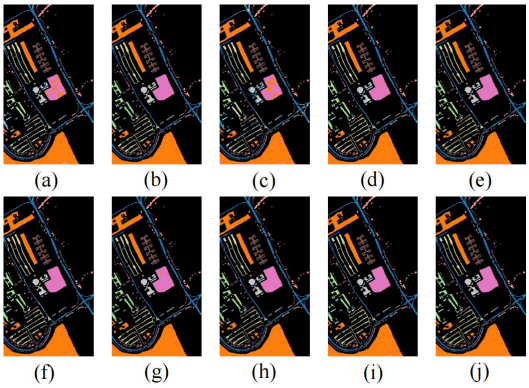


Fig. 11. Classification maps of different methods on the PaviaU dataset. (a) HybridSN. (b) SPRN. (c) SpectralFormer. (d) SSFTT. (e) GAHT. (f) MorphFormer. (g) GSC-ViT. (h) CAEVT. (i) MsGsFNet. (j) ASSCA-Net.

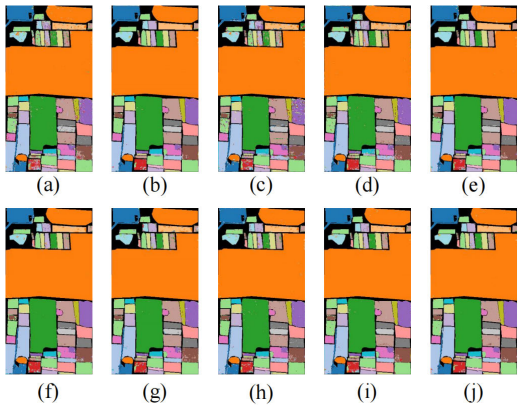


Fig. 12. Classification maps of different methods on the WHU-Hi-HongHu dataset. (a) HybridSN. (b) SPRN. (c) SpectralFormer. (d) SSFTT. (e) GAHT. (f) MorphFormer. (g) GSC-ViT. (h) CAEVT. (i) MsGsFNet. (j) ASSCA-Net.

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TABLE V

QUANTITATIVE CLASSIFICATION RESULTS ON THE SALINAS DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	HybridSN	SPRN	spectralformer	SSFTT	GAHT	morphformer	GSC-ViT	CAEVT	MsGsFNet	ASSCA-Net
C1	98.90±0.66	99.47±1.58	94.04±1.22	99.77±0.42	99.69±0.45	98.61±2.35	98.70±2.75	99.92±0.13	**99.94±0.10**	**99.94±0.17**
C2	99.72±0.22	**100.00±0.00**	98.03±2.23	99.71±0.33	99.67±0.47	99.72±0.43	99.51±0.86	99.83±0.25	**100.00±0.00**	**100.00±0.00**
C3	95.81±4.14	**99.76±0.27**	90.08±6.16	92.40±9.71	98.93±0.98	95.05±7.02	96.20±7.10	95.83±3.52	99.79±0.26	99.49±1.00
C4	97.56±2.11	**99.46±0.56**	94.44±2.10	98.78±1.34	99.12±1.35	99.21±1.05	99.07±1.34	98.39±2.41	98.54±1.24	99.37±1.11
C5	93.98±2.47	97.30±2.18	84.14±4.79	97.76±1.09	**97.96±1.72**	94.53±3.17	95.72±3.04	93.81±3.27	97.68±2.13	97.76±1.79
C6	99.99±0.01	**100.00±0.00**	99.70±0.52	99.91±0.25	99.97±0.06	99.89±0.33	99.98±0.05	99.96±0.08	**100.00±0.00**	99.93±0.17
C7	99.39±0.63	**99.98±0.02**	97.74±1.45	99.60±0.54	99.87±0.21	99.91±0.10	99.72±0.39	99.78±0.41	99.88±0.11	99.84±0.31
C8	87.89±2.76	94.92±2.07	82.64±2.27	90.66±2.23	92.87±2.00	93.02±1.76	92.36±3.46	90.17±2.48	94.39±2.25	**97.96±1.18**
C9	99.27±0.54	**100.00±0.00**	97.81±0.92	98.86±1.38	99.75±0.40	99.59±0.42	**100.00±0.00**	99.37±1.01	**100.00±0.00**	99.96±0.08
C10	93.06±3.02	98.36±0.72	88.35±3.74	95.85±1.67	97.83±1.18	96.93±1.79	97.98±0.75	96.37±1.74	97.92±0.94	**99.11±1.36**
C11	84.60±9.16	95.64±6.14	71.94±13.03	89.00±13.70	94.88±7.15	87.77±10.35	93.81±8.36	89.26±10.21	92.36±8.91	**99.57±0.34**
C12	98.64±1.19	99.49±0.53	96.52±2.52	99.49±0.48	99.92±0.18	99.41±0.82	98.91±1.11	99.12±1.29	99.69±0.55	**99.97±0.06**
C13	99.38±0.80	99.87±0.22	98.33±2.24	99.86±0.25	99.79±0.32	99.79±0.24	99.25±1.37	99.69±0.58	99.98±0.07	**100.00±0.00**
C14	98.88±1.31	99.35±0.73	98.59±1.36	95.83±4.59	98.88±1.45	98.80±0.93	99.15±1.83	98.17±1.26	**99.72±0.31**	98.69±1.14
C15	77.45±2.36	93.45±2.49	78.04±3.43	82.92±3.94	89.05±2.17	88.94±3.25	91.05±4.96	87.39±3.32	92.75±3.33	**96.91±1.70**
C16	94.96±5.20	98.79±0.99	91.23±3.48	96.90±2.29	96.82±2.34	97.71±1.76	98.08±1.69	96.49±4.35	98.48±0.48	**99.12±1.82**
OA(%)	92.78±0.74	97.47±0.36	89.64±1.09	94.47±0.66	96.44±0.50	95.91±0.47	96.35±0.39	95.03±0.61	97.34±0.57	**98.87±0.32**
AA(%)	94.97±1.00	98.36±0.57	91.35±1.34	96.08±1.03	97.81±0.50	96.80±0.89	97.47±0.82	96.47±0.88	98.20±0.56	**99.23±0.17**
κ *100	91.96±0.82	97.18±0.41	88.47±1.21	93.85±0.73	96.04±0.56	95.44±0.53	95.94±0.43	94.47±0.68	97.04±0.64	**98.75±0.35**

TABLE VI

QUANTITATIVE CLASSIFICATION RESULTS ON THE PAVIAU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	HybridSN	SPRN	spectralformer	SSFTT	GAHT	morphformer	GSC-ViT	CAEVT	MsGsFNet	ASSCA-Net
C1	94.09±2.05	98.45±0.85	86.57±2.25	96.38±1.71	97.88±1.06	96.95±1.35	98.05±0.78	97.91±0.69	98.21±0.67	**98.81±0.67**
C2	96.44±1.24	99.59±0.41	97.25±0.85	99.11±0.37	99.41±0.32	99.55±0.21	99.18±0.69	99.42±0.29	99.61±0.19	**99.88±0.08**
C3	69.16±7.14	92.49±4.67	56.05±16.66	86.11±6.77	84.64±4.63	88.20±4.31	91.27±4.93	87.04±3.42	87.36±6.77	**95.23±2.45**
C4	93.08±3.00	95.36±1.68	91.23±2.42	**97.14±1.88**	97.02±0.94	95.51±2.05	96.12±1.69	97.08±1.59	96.09±1.60	96.48±1.27
C5	**99.97±0.07**	99.88±0.23	99.95±0.16	99.72±0.37	99.87±0.27	99.64±0.92	99.84±0.22	99.67±0.45	99.92±0.18	99.20±1.66
C6	79.38±7.48	**99.22±0.63**	77.52±3.75	97.88±1.18	97.91±1.84	99.15±1.04	97.68±2.25	98.35±1.35	97.89±1.95	98.67±1.19
C7	72.38±7.26	94.57±5.52	47.75±12.45	84.44±4.93	88.83±7.36	90.16±6.56	93.43±5.73	88.36±6.08	88.27±8.73	**99.18±1.22**
C8	88.14±5.32	95.12±2.68	74.06±6.86	92.88±3.21	93.57±2.99	93.62±2.70	96.08±2.52	93.62±4.14	96.06±1.30	**96.33±1.96**
C9	96.15±5.41	98.27±1.29	90.29±5.31	97.24±3.55	97.53±1.00	96.30±4.83	97.65±2.65	**99.49±0.56**	97.45±1.58	94.45±2.78
OA(%)	91.13±1.47	98.16±0.42	87.22±0.94	94.47±0.66	97.24±0.47	97.38±0.47	97.76±0.54	97.45±0.46	96.75±0.31	**98.63±0.25**
AA(%)	87.64±2.23	97.00±0.77	80.07±2.09	94.55±0.77	95.18±1.27	95.45±0.90	96.59±0.83	95.66±1.05	95.65±1.23	**97.58±0.50**
κ *100	88.17±2.00	97.56±0.55	82.90±1.31	95.69±0.40	96.34±0.63	96.53±0.62	97.03±0.71	96.62±0.61	96.87±0.61	**98.18±0.33**

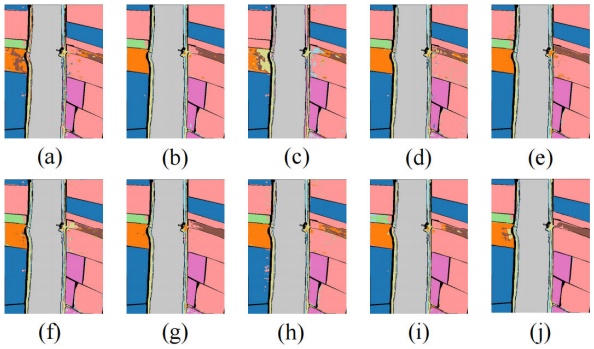


Fig. 13. Classification maps of different methods on the WHU-Hi-LongKou dataset. (a) HybridSN. (b) SPRN. (c) SpectralFormer. (d) SSFTT. (e) GAHT. (f) MorphFormer. (g) GSC-ViT. (h) CAEVT. (i) MsGsFNet. (j) ASSCA-Net.

AA, and Kappa, and leads in accuracy for most classes. Some compared methods perform well on a few individual classes, but their overall performance remains inferior to the proposed method. Pure Transformers lack sufficient adaptability to local spatial–spectral structures under limited samples. Hybrid CNN–Transformer methods alleviate this to some extent, but still fall short of ASSCA-Net in the collaborative design of differentiated band perception, spatial modeling of irregular ground objects, and deep spatial–spectral interaction.

As shown in Table VI and Fig. 11, this urban scene contains 9 land-cover classes. ASSCA-Net achieves the best results on all three overall metrics and the highest accuracy for most classes. HybridSN is limited by local convolutional receptive fields and tends to confuse spectrally similar classes. SpectralFormer can model long-range dependencies but lacks adaptive spatial representation for strip-like objects, limiting the learning of discriminative features. These results indicate that a single CNN or Transformer paradigm struggles to form stable competitiveness in complex urban scenes.

As shown in Fig. 12 and Table VII, this high-resolution UAV image contains 22 fine-grained agricultural classes. ASSCA-Net achieves higher OA, AA, and Kappa than all compared methods, and obtains the best results for the vast majority of classes. Only a few individual classes are slightly bettered by SPRN and others. Some pure CNN or single Transformer models exhibit lower accuracy and larger fluctuations under high class diversity, indicating insufficient generalization and stability. In contrast, the anchor-driven fine-grained spectral modeling and multiscale strip spatial extraction make ASSCA-Net's advantages more prominent in fine-grained classification.

As shown in Fig. 13 and Table VIII, this scene contains nine land-cover classes with interclass spectral similarity and sample imbalance. ASSCA-Net leads in all three metrics and achieves the best results on multiple difficult classes. GSC-ViT, which also belongs to the spatial–spectral joint modeling framework, is the closest competitor. Traditional CNN or pure Transformer methods tend to suffer from accuracy fluctuations and larger standard deviations in regions with spectral similarity and class imbalance. In contrast,

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TABLE VII

QUANTITATIVE CLASSIFICATION RESULTS ON THE WHU-HI-HONGHU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	HybridSN	SPRN	spectralformer	SSFTT	GAHT	morphformer	GSC-ViT	CAEVT	MsGsFNet	ASSCA-Net
C1	96.99±0.54	98.45±0.72	97.35±0.93	97.98±1.07	98.26±0.71	98.30±0.65	98.23±1.06	97.95±0.58	98.43±0.65	**98.78±0.60**
C2	85.34±3.52	**92.45±3.25**	82.43±3.39	87.09±3.27	91.38±2.76	89.68±2.69	89.05±5.81	86.64±3.20	92.40±2.21	91.32±2.08
C3	94.33±0.97	97.35±1.02	93.89±1.07	95.26±1.36	96.74±0.70	96.97±0.92	96.38±1.57	96.52±1.26	97.23±0.51	**97.54±0.61**
C4	99.57±0.11	99.74±0.09	99.40±0.17	99.64±0.15	99.71±0.10	99.70±0.21	99.60±0.15	99.68±0.11	99.78±0.06	**99.80±0.08**
C5	83.02±2.26	92.81±2.81	85.34±2.73	85.69±3.36	91.70±1.42	90.45±2.34	88.57±4.53	86.55±6.19	91.19±1.56	**94.30±1.37**
C6	97.34±0.39	99.06±0.31	97.46±0.38	98.81±0.38	99.04±0.21	98.91±0.41	98.60±0.64	98.36±0.58	99.09±0.31	**99.32±0.21**
C7	90.48±1.74	95.71±1.50	89.61±1.14	91.63±1.79	96.05±1.14	95.88±1.41	94.98±1.87	93.91±1.25	95.87±1.19	**96.57±1.26**
C8	53.43±5.25	79.99±4.31	54.69±2.56	72.52±3.43	77.06±3.57	80.48±2.75	73.50±5.49	72.45±3.44	74.14±3.28	**84.94±2.32**
C9	98.50±0.37	99.48±0.14	98.67±0.39	98.52±0.81	99.19±0.37	99.16±0.34	98.90±0.52	98.77±0.28	99.33±0.32	**99.53±0.20**
C10	87.50±1.63	95.41±1.38	75.44±1.57	92.81±1.84	95.85±1.19	95.84±1.51	95.26±1.72	94.24±1.85	95.51±1.77	**96.35±0.98**
C11	82.39±3.21	93.96±1.15	80.45±2.08	89.12±2.78	94.29±1.88	94.78±1.24	91.91±1.20	90.06±1.92	93.60±1.34	**96.06±0.78**
C12	77.57±3.83	90.19±2.30	73.87±2.22	82.12±3.89	89.09±1.70	88.08±2.67	86.61±2.98	85.88±1.68	86.93±2.23	**92.91±1.11**
C13	87.58±1.17	95.37±0.79	87.10±1.21	89.60±2.04	95.06±1.15	94.14±1.55	92.42±2.42	93.04±1.04	94.86±0.58	**96.21±0.41**
C14	92.62±1.01	**97.07±1.14**	89.82±2.10	94.21±2.29	96.40±0.69	96.02±1.78	96.32±1.23	94.75±1.84	96.07±1.32	96.68±0.88
C15	72.86±9.54	**95.98±2.82**	83.34±4.39	84.14±5.95	92.24±3.94	89.39±8.78	89.28±5.95	90.43±3.37	91.33±4.14	93.53±3.62
C16	95.07±1.83	98.22±0.90	95.72±1.85	95.95±2.39	97.58±1.18	97.96±0.81	98.31±1.02	97.22±1.44	98.07±0.84	**98.84±0.87**
C17	86.24±4.86	95.78±3.45	82.13±4.77	92.51±3.73	95.40±2.35	91.95±7.02	91.92±2.68	93.75±2.76	95.57±1.83	**97.01±1.12**
C18	79.45±3.90	94.72±2.32	84.03±2.70	88.65±5.57	93.12±2.56	94.92±1.81	91.69±3.81	91.84±2.50	92.44±2.35	**95.41±1.94**
C19	92.27±1.32	95.88±1.06	91.86±1.95	92.91±2.27	96.28±0.97	94.98±1.20	94.88±1.49	94.52±1.09	95.74±1.23	**96.54±1.03**
C20	89.76±1.54	96.45±0.78	90.60±1.74	89.74±2.85	94.63±1.08	95.23±1.22	93.26±1.51	93.37±1.90	96.01±0.69	**97.15±0.85**
C21	44.82±12.2	79.87±6.24	48.87±11.13	71.64±10.85	79.25±7.20	82.13±8.20	70.38±14.43	70.96±9.54	81.17±7.20	**91.22±4.18**
C22	82.79±3.87	91.35±4.03	86.89±3.09	87.94±3.51	93.24±3.80	92.81±3.30	91.76±3.48	89.29±5.05	92.37±3.91	**94.03±3.80**
OA(%)	94.19±0.28	97.62±0.21	93.59±0.19	95.75±0.30	97.46±0.11	97.34±0.17	96.74±0.24	96.48±0.27	97.40±0.57	**98.12±0.11**
AA(%)	85.00±1.18	94.33±0.40	84.95±0.63	89.93±1.00	93.71±0.31	93.53±0.57	91.90±0.84	91.37±0.65	93.51±0.56	**95.64±0.35**
κ *100	92.65±0.35	97.00±0.26	91.89±0.24	94.63±0.38	96.79±0.14	96.64±0.22	95.88±0.30	95.54±0.34	96.71±0.64	**97.63±0.14**

TABLE VIII

QUANTITATIVE CLASSIFICATION RESULTS ON THE WHU-HI-LONGKOU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	HybridSN	SPRN	spectralformer	SSFTT	GAHT	morphformer	GSC-ViT	CAEVT	MsGsFNet	ASSCA-Net
C1	97.97±1.26	**99.71±0.25**	98.91±1.32	99.25±0.76	99.68±0.34	99.25±0.51	99.53±0.51	99.36±0.64	99.69±0.31	98.08±0.24
C2	65.54±13.8	92.69±4.56	64.96±16.22	92.12±6.09	92.43±6.52	90.85±5.44	89.59±10.31	92.04±8.07	94.23±4.53	**94.25±8.70**
C3	81.76±4.65	84.05±7.78	56.63±31.36	71.74±18.16	81.05±6.53	80.24±8.74	**87.18±6.59**	81.39±6.63	46.46±38.41	81.95±7.45
C4	96.72±1.20	98.82±0.51	95.65±0.77	97.34±1.72	98.15±0.87	97.48±1.39	98.36±0.92	98.01±0.92	98.49±0.75	**99.95±1.26**
C5	33.88±16.0	79.22±18.87	26.99±15.51	74.73±18.31	79.58±14.8	75.54±16.33	**85.31±5.41**	77.37±15.81	37.36±25.56	73.23±17.93
C6	95.00±2.84	98.48±1.74	91.32±4.07	96.26±2.00	98.77±0.98	96.01±3.35	97.72±1.85	97.57±1.72	94.53±6.01	**99.37±3.52**
C7	**99.99±0.02**	99.88±0.20	99.87±0.15	99.94±0.08	99.79±0.35	99.93±0.11	99.88±0.16	99.95±0.08	99.95±0.10	99.95±0.24
C8	81.26±8.19	90.21±5.00	68.76±9.36	91.20±3.72	88.37±6.57	90.27±6.00	89.16±5.03	88.77±4.15	87.05±4.85	**95.98±4.10**
C9	57.29±10.1	87.50±6.30	27.74±9.74	82.88±9.00	**87.66±3.92**	72.19±7.80	84.26±6.77	79.77±7.20	73.25±14.35	86.85±3.92
OA(%)	93.58±0.89	97.84±0.36	91.43±0.96	96.82±0.44	97.51±0.33	96.63±0.60	97.55±0.70	97.15±0.52	95.71±0.99	**98.08±0.53**
AA(%)	78.82±3.10	92.28±2.47	70.09±4.50	89.50±3.27	91.72±1.78	89.08±1.74	92.33±1.80	90.47±2.03	81.22±5.30	**92.87±2.55**
κ *100	91.53±1.18	97.16±0.48	88.69±1.28	95.82±0.58	96.72±0.43	95.56±0.78	96.78±0.92	96.25±0.68	94.33±1.32	**97.47±0.70**

ASSCA-Net maintains high accuracy while exhibiting smaller standard deviations, demonstrating good stability and reproducibility.

In summary, on four public datasets with different scenes and resolutions, ASSCA-Net achieves the best or comparable-to-best overall performance. Its classification maps are superior to most compared methods in terms of spatial consistency and boundary delineation. Relying solely on CNNs or Transformers is insufficient to fully exploit the joint spatial–spectral representation of HSIs. The proposed hierarchical architecture systematically compensates for these limitations from a design perspective, validating the effectiveness and generalization capability of the proposed method.

D. Parameter Experiments

1) Patch Size Sensitivity Experiment: With the regularization coefficient λ fixed at the optimal value for each dataset, the patch size is varied from 6×6 to 15×15. Each configuration is run independently 10 times. Fig. 14 presents the mean $\pm$ standard deviation curves of OA, AA, and Kappa ($\times 100$) against patch size.

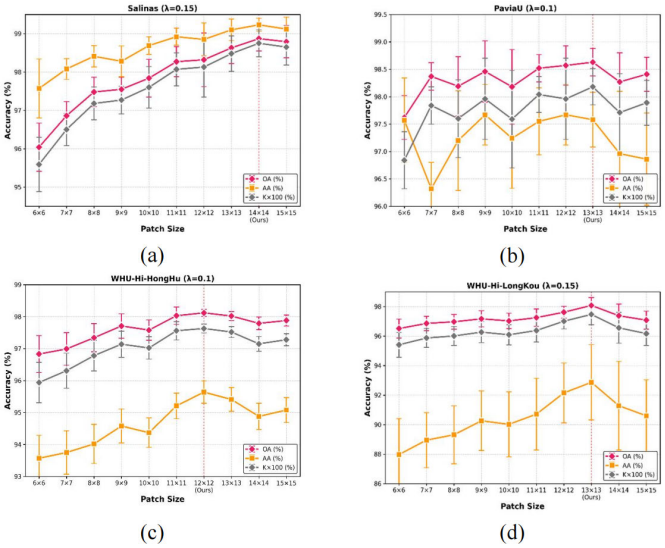


Fig. 14. Impact of patch size on OA, AA, and Kappa. (a) Salinas dataset. (b) Pavia University dataset. (c) WHU-Hi-HongHu dataset. (d) WHU-Hi-LongKou dataset.

The experimental results show that the optimal patch size varies by dataset, but generally falls in the range of 12×12 to 14×14. A moderately large window helps capture sufficient spatial context and improves discriminability. However, an excessively large window introduces redundant background information and boundary effects, which instead degrades performance. The optimal values for each dataset are as follows: 13×13 for PaviaU and WHU-Hi-LongKou, 12×12

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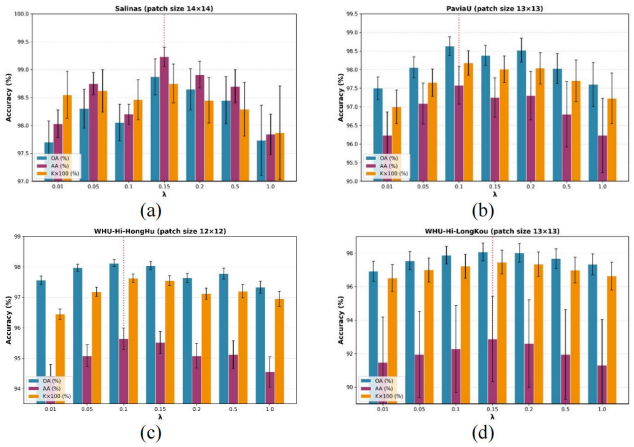


Fig. 15. Impact of λ on OA, AA, and Kappa. (a) Salinas dataset. (b) Pavia University dataset. (c) WHU-Hi-HongHu dataset. (d) WHU-Hi-LongKou dataset.

for WHU-Hi-HongHu, and 14×14 for Salinas. This difference is related to the spatial resolution and object scales of each dataset. WHU-Hi-HongHu has the highest resolution, so a smaller window can already capture complete object structures. Salinas, with lower resolution, requires a larger window for sufficient spatial context.

2) Regularization Weight λ Sensitivity Experiment: With the optimal patch size fixed for each dataset, λ is varied within {0.01, 0.05, 0.10, 0.15, 0.20, 0.50, 1.00}. Each configuration is run independently 10 times. Fig. 15 presents the bar charts of mean $\pm$ standard deviation of OA, AA, and Kappa ($\times100$) against λ.

The results indicate that λ controls the strength of the intraclass spectral variance regularization. It is found that the optimal λ for each dataset concentrates in the range of 0.10–0.15: 0.10 for PaviaU and WHU-Hi-HongHu, and 0.15 for WHU-Hi-LongKou and Salinas. When λ is too small, the regularization constraint is insufficient, and intraclass spectral features are dispersed, making it difficult to form compact class centers. When λ is too large, the regularization term dominates the loss function excessively, forcing the network to overly compress spectral diversity, which in turn weakens feature discriminability. A moderate λ balances classification error and feature compactness, thus achieving the best performance.

3) Other Hyperparameters: Besides patch size and λ, ASSCA-Net involves several other hyperparameters, including σ and γ in SSAM, β in the ASAM cost matrix, ε in the optimal transport formulation, and τ in the ASAM gating. These hyperparameters are not highly sensitive within reasonable ranges.

Their default values were determined through grid search on the validation sets of all four datasets, with candidate ranges of $\sigma \in \{0.5, 1.0, 2.0\}$, $\gamma \in \{1.0, 2.0, 5.0\}$, $\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, $\varepsilon \in \{0.01, 0.05, 0.1, 0.5\}$, and $\tau \in \{0.2, 0.5, 1.0\}$. The configuration achieving the highest average validation OA across the four datasets was selected as the default, with $\sigma = 1.0$, $\gamma = 2.0$, $\beta = 0.5$, $\varepsilon = 0.1$, and $\tau = 0.5$. These values are then fixed across all datasets to ensure a fair and unified evaluation protocol, with λ being the only dataset-dependent hyperparameter whose sensitivity is analyzed in Fig. 15.

TABLE IX
MODULE ABLATION RESULTS ON THE SALINAS DATASET

Configuration	OA(%)	AA(%)	κ*100
Complete Model	98.87±0.32	99.23±0.17	98.75±0.35
Remove SSAM	95.34±0.43	96.16±0.31	94.91±0.49
Replace ASAM	95.17±0.47	96.08±0.36	94.62±0.53
Replace DMSP	95.49±0.40	96.21±0.28	95.06±0.46
Replace ASAM+ DMSP	91.83±0.73	92.91±0.58	91.39±0.86
Replace BCAF with Concat	94.52±0.56	96.11±0.39	95.18±0.51
Replace BCAF with Element-wise Sum	95.06±0.48	96.54±0.35	95.61±0.45
Replace BCAF with Gated Fusion	95.48±0.44	96.82±0.32	96.02±0.41
Replace BCAF with SE Fusion	95.89±0.38	97.11±0.29	96.43±0.36

TABLE X
MODULE ABLATION RESULTS ON THE PAVIAU DATASET

Configuration	OA(%)	AA(%)	κ*100
Complete Model	98.63±0.25	97.58±0.50	98.18±0.33
Remove SSAM	95.86±0.42	94.94±0.67	95.61±0.53
Replace ASAM	95.24±0.51	93.87±0.74	95.13±0.64
Replace DMSP	96.14±0.38	95.06±0.62	95.57±0.47
Replace ASAM+ DMSP	93.46±0.68	91.89±0.85	93.16±0.79
Replace BCAF with Concat	95.47±0.46	94.63±0.71	95.32±0.58
Replace BCAF with Element-wise Sum	95.96±0.40	95.18±0.63	95.78±0.52
Replace BCAF with Gated Fusion	96.31±0.35	95.52±0.57	96.12±0.47
Replace BCAF with SE Fusion	96.68±0.31	95.89±0.50	96.48±0.40

E. Ablation Experiments

To verify the effectiveness of each module, ablation experiments are conducted across the four datasets. The complete model ASSCA-Net contains all four modules: SSAM, ASAM, DMSP, and BCAF. The ablation settings are as follows.

1) *Remove SSAM:* The original input is directly compressed by a 1×1 convolution and fed into the dual branches without spectral-guided spatial aggregation.
2) *Replace ASAM:* The anchor-driven spectral attention is replaced with global channel attention, where global average pooling compresses the spatial dimension, followed by a fully connected layer and Sigmoid to generate channel-wise weights.
3) *Replace DMSP:* The DMSP is replaced with a single-scale 3×3 depthwise convolution for local spatial feature extraction.
4) *Replace BCAF:* The BCAF is replaced with four alternative fusion strategies, including simple concatenation, element-wise sum, gated fusion, and SE fusion.

The experimental results are shown in Tables IX–XII.

SSAM serves as the network entry. Through spectral-guided spatial aggregation, it suppresses background noise and enhances intraclass spectral consistency. After its removal, accuracy drops noticeably across all datasets. This shows that the high-quality input it provides for subsequent branches is indispensable.

Replacing ASAM with global channel attention leads to a more significant performance loss. Channel attention shares

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TABLE XI
MODULE ABLATION RESULTS ON THE WHU-HI-HONGHU DATASET

Configuration	OA(%)	AA(%)	κ*100
Complete Model	98.12±0.11	95.64±0.35	97.63±0.14
Remove SSAM	96.27±0.28	93.46±0.52	95.07±0.37
Replace ASAM	94.86±0.39	92.37±0.63	93.94±0.48
Replace DMSP	94.53±0.44	91.96±0.69	93.51±0.41
Replace ASAM+ DMSP	91.37±0.61	89.74±0.87	90.86±0.72
Replace BCAF with Concat	95.16±0.34	93.17±0.58	94.52±0.43
Replace BCAF with Element-wise Sum	95.62±0.29	93.68±0.52	94.94±0.38
Replace BCAF with Gated Fusion	95.94±0.25	94.02±0.47	95.27±0.34
Replace BCAF with SE Fusion	96.31±0.21	94.41±0.41	95.68±0.29

TABLE XII
MODULE ABLATION RESULTS ON THE WHU-HI-LONGKOU DATASET

Configuration	OA(%)	AA(%)	κ*100
Complete Model	98.06±0.53	92.87±2.55	97.47±0.70
Remove SSAM	94.97±0.67	89.96±2.74	93.54±0.87
Replace ASAM	93.47±0.84	87.84±3.02	92.46±1.07
Replace DMSP	94.86±0.64	89.47±2.68	94.04±0.81
Replace ASAM+ DMSP	90.67±0.97	84.86±3.19	89.74±1.14
Replace BCAF with Concat	93.86±0.76	88.24±2.88	92.93±0.94
Replace BCAF with Element-wise Sum	94.31±0.68	88.92±2.74	93.41±0.86
Replace BCAF with Gated Fusion	94.68±0.61	89.43±2.62	93.78±0.78
Replace BCAF with SE Fusion	95.02±0.54	89.96±2.48	94.16±0.70

the same set of weights across all pixels. It cannot reflect spatially varying spectral discriminative differences. In contrast, ASAM uses learnable anchors and optimal transport to adaptively match a spectral prototype for each pixel. This makes same-class spectra more clustered in the embedding space and is more suitable for fine-grained classification.

Replacing DMSP with a fixed 3×3 convolution causes a clear decline, especially on scenes with strip-like or winding objects. Standard convolution struggles to capture long-range context along object orientations. DMSP overcomes this limitation through gradient-guided direction selection and multiscale strip pooling.

When ASAM and DMSP are jointly replaced, the performance degradation exceeds the sum of their individual drops, indicating that spectral representation and directional spatial modeling are functionally complementary.

Replacing BCAF with simple concatenation weakens the fusion gain, as concatenation merely stacks features without semantic interaction. In contrast, BCAF enables bidirectional cross-attention, allowing spectral and spatial features to mutually enhance each other. This yields synergistic improvement beyond linear fusion.

In summary, SSAM, ASAM, DMSP, and BCAF perform front-end preprocessing, pixelwise spectral modeling, directional context extraction, and cross-modal fusion, respectively. Their roles are clearly defined and sequentially connected. The complete model achieves the best performance across all datasets, while removing any component leads to degradation, quantitatively verifying the necessity of each module.

TABLE XIII
COMPARISON OF PARAMETERS, FLOPS, TRAINING TIME, AND TESTING TIME OF DIFFERENT MODELS

Method	Params (M)	FLOPs (G)	Train Time (min)	Test Time (s)
HybridSN	8.26	2.48	14.2	8.5
SPRN	0.18	0.012	2.4	0.8
spectralformer	0.14	0.032	3.2	1.2
SSFTT	0.38	0.020	3.8	1.3
GAHT	0.76	0.049	5.0	2.2
morphformer	0.10	0.008	2.0	0.7
GSC-ViT	0.17	0.010	2.5	0.9
CAEVT	0.36	0.12	6.0	4.0
MsGsFNet	0.30	0.026	3.5	1.2
ASSCA-Net	0.32	0.018	3.2	1.5

F. Computational Efficiency Analysis

To evaluate the proposed method from an efficiency perspective, all models are assessed under identical input configurations to ensure a fair comparison. Specifically, experiments are conducted on the Pavia University dataset with a patch size of 13×13, yielding an input shape of (1, 1, 103, 13, and 13). This configuration aligns with the mainstream settings adopted in the comparative experiments, thereby enabling an equitable reflection of the computational efficiency disparities among the evaluated models. The number of parameters (Params), floating-point operations (FLOPs), training time, and test time are recorded for each model, as summarized in Table XIII.

As evident from Table XIII, HybridSN—a representative 3-D CNN-based method—exhibits the largest parameter count and FLOPs, along with the longest training and testing durations; yet, its classification accuracy among the compared methods remains unremarkable. Lightweight approaches, such as SPRN, MorphFormer, and GSC-ViT, possess parameter counts below 0.2M and demonstrate notable efficiency advantages. However, their overall accuracies are substantially inferior to those achieved by ASSCA-Net, particularly on the Salinas and WHU-Hi-HongHu datasets.

In contrast, ASSCA-Net maintains a parameter count of 0.32M, FLOPs of 0.018G, a training time of 3.2 min, and a test time of 1.5 s—values that reside within the same order of magnitude as those of MsGsFNet and SSFTT. Crucially, ASSCA-Net attains the highest classification accuracy across all four datasets. In other words, ASSCA-Net does not pursue extreme lightweight design at the expense of performance; rather, it achieves a substantial improvement in classification accuracy with only a modest increase in computational overhead. Considering the tradeoff between accuracy and efficiency, ASSCA-Net strikes a favorable balance.

G. t-SNE Feature Visualization

To further analyze the discriminative ability of the proposed method at the feature level, feature visualization is carried out on the optimal models after training on the four datasets. The features from the layer before the classifier are extracted for both ASSCA-Net and GSC-ViT. t-distributed stochastic neighbor embedding (t-SNE) is applied to reduce the features to a 2-D plane. The scatter points are colored by class. Fig.

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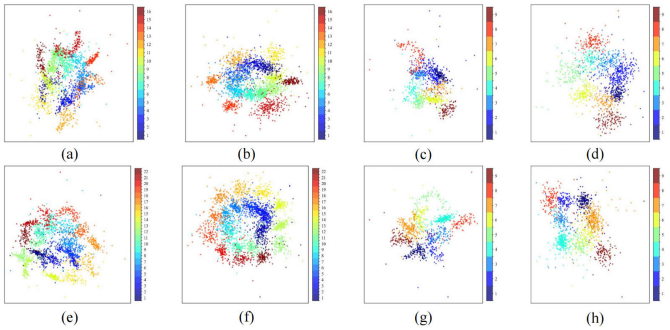


Fig. 16. t-SNE feature visualization on the GSC-ViT and ASSCA-Net. (a) and (b) Salinas dataset. (c) and (d) PaviaU dataset. (e) and (f) WHU-Hi-HongHu dataset. (g) and (h) WHU-Hi-LongKou dataset.

16 shows the distribution comparison for each dataset. The color bar on the right side of each subfigure indicates the class numbers.

From the visualization results, in the feature distribution of GSC-ViT, the scatter points of different classes overlap clearly. The interclass boundaries are blurred. Samples of the same class show a relatively high degree of dispersion. The confusion is particularly serious among ground object classes that are spectrally similar or spatially adjacent. In contrast, in the feature distribution of ASSCA-Net, samples of the same class cluster significantly together. The boundaries between different classes are clear. Interclass confusion is greatly reduced. The scatter points of the same class form more compact and coherent distribution patterns. This advantage is especially prominent on the WHU-Hi-HongHu dataset, which has the largest number of classes at 22.

The above visualization results are highly consistent with the quantitative metrics OA, AA, and Kappa in the comparative experiments. They further verify, from the feature representation level, the effectiveness of ASSCA-Net in enhancing intraclass compactness and interclass separability.

IV. Conclusion

Targeting the problems of diverse ground object morphology, significant spectral variability, and insufficient spatial–spectral feature interaction in HSI classification, this article proposes an ASSCA-Net. The network constructs a hierarchical spatial–spectral collaborative framework with four functional modules. The SSAM performs weighted neighborhood aggregation at the input stage based on SAS. This suppresses noise and improves intraclass spectral consistency. The ASAM module assigns a spectral prototype to each pixel via learnable anchors and optimal transport. This enables object-level differentiated band perception. The DMSP module adaptively adjusts the strip orientation according to the local gradient direction. This characterizes the long-range spatial dependencies of strip-like and winding ground objects. The BCAF module promotes query-based interaction and adaptive fusion of spectral and spatial features through bidirectional cross-attention.

Experimental results on four public datasets, Salinas, Pavia University, WHU-Hi-HongHu, and WHU-Hi-LongKou, demonstrate that ASSCA-Net outperforms mainstream compared methods in OA, AA, and Kappa coefficient. The ablation study verifies the individual contribution and synergistic complementarity of each module. The parameter sensitivity analysis provides a reference for selecting the patch size and the regularization coefficient. The t-SNE visualization further shows that the learned representations possess stronger intraclass compactness and interclass separability.

Based on the above spatial–spectral collaborative learning paradigm, future research can proceed in two directions. First, enhance the model robustness under limited labeled conditions, so that it maintains high discriminative ability in few-sample scenarios. This addresses the label scarcity problem in real hyperspectral mapping. Second, explore the transfer potential of the learned spatial–spectral joint representation to related tasks, such as object detection and multitemporal change analysis. The principle of spectral–spatial collaborative modeling is expected to remain effective without fundamental architectural modifications.

References

[1] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[2] M. Ahmad et al., "A comprehensive survey for hyperspectral image classification: The evolution from conventional to transformers and mamba models," *Neurocomputing*, vol. 644, Sep. 2025, Art. no. 130428.

[3] R. G. Vaughan, W. M. Calvin, and J. V. Taranik, "SEBASS hyperspectral thermal infrared data: Surface emissivity measurement and mineral mapping," *Remote Sens. Environ.*, vol. 85, no. 1, pp. 48–63, Apr. 2003.

[4] C. Zhang, M. Yi, F. Ye, Q. Xu, X. Li, and Q. Gan, "Application and evaluation of deep neural networks for airborne hyperspectral remote sensing mineral mapping: A case study of the baiyanghe uranium deposit in Northwestern Xinjiang, China," *Remote Sens.*, vol. 14, no. 20, p. 5122, Oct. 2022.

[5] H. Zeng, X. Han, and Q. Liu, "Mineral detection from hyperspectral images using a spatial–spectral residual convolution neural network," *J. Phys., Conf. Ser.*, vol. 1894, no. 1, Apr. 2021, Art. no. 012104.

[6] S. Ofori-Ampofo, C. Pelletier, and S. Lang, "Crop type mapping from optical and radar time series using attention-based deep learning," *Remote Sens.*, vol. 13, no. 22, p. 4668, Nov. 2021.

[7] P. S. Thenkabail, R. B. Smith, and E. De Pauw, "Hyperspectral vegetation indices and their relationships with agricultural crop characteristics," *Remote Sens. Environ.*, vol. 71, no. 2, pp. 158–182, Feb. 2000.

[8] C. Nguyen, V. Sagan, J. Skobalski, and J. I. Severo, "Early detection of wheat yellow rust disease and its impact on terminal yield with multi-spectral UAV-imagery," *Remote Sens.*, vol. 15, no. 13, p. 3301, Jun. 2023.

[9] F. Zeng et al., "Monitoring inland water via sentinel satellite constellation: A review and perspective," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 340–361, Oct. 2023.

[10] Y. Zhang, L. Wu, L. Deng, and B. Ouyang, "Retrieval of water quality parameters from hyperspectral images using a hybrid feedback deep factorization machine model," *Water Res.*, vol. 204, Oct. 2021, Art. no. 117618.

[11] Y. Zhang et al., "Mapping water quality parameters in urban rivers from hyperspectral images using a new self-adapting selection of multiple artificial neural networks," *Remote Sensing*, vol. 12, no. 2, p. 336, Jan. 2020.

[12] Y. Zhang, J. Zhu, and K. Cao, "Few-shot military target detection with text-to-image and vision-language models," in *Proc. 7th Int. Conf. Image Graph. Process.*, Jan. 2024, pp. 47–53.

[13] M. Shimoni, R. Haelterman, and C. Perneel, "Hyperspectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag., Replaces Newslett.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[14] Y. Song et al., "Deep learning for hyperspectral image classification: A comprehensive review and future predictions," *Inf. Fusion*, vol. 123, Nov. 2025, Art. no. 103285.

[15] J. Bai, E. Zhang, L. Yang, X. Li, and S. Zhang, "Generative adversarial network-based spectral–spatial feature learning for hyperspectral image classification," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–8.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:16:03 UTC from IEEE Xplore. Restrictions apply.

[16] C. Yang and D. Xue, "Spectral attention graph convolutional network for hyperspectral image classification," in *Proc. 6th Int. Conf. Commun., Inf. Syst. Comput. Eng. (CISCE)*, May 2024, pp. 687–692.

[17] X. Wang, J. Liu, W. Chi, W. Wang, and Y. Ni, "Advances in hyperspectral image classification methods with small samples: A review," *Remote Sens.*, vol. 15, no. 15, p. 3795, Jul. 2023.

[18] X. Wang and Y. Fan, "Multiscale densely connected attention network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1617–1628, 2022.

[19] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[20] K. Wu, J. Fan, P. Ye, and M. Zhu, "Hyperspectral image classification using spectral–spatial token enhanced transformer with hash-based positional embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507016.

[21] Z. Zheng, M. Debbagh, X. Zhou, S. Sun, and Y. Huang, "Graph-transformer with spatial–spectral features fusion for hyperspectral image classification," *Expert Syst. Appl.*, vol. 264, Mar. 2025, Art. no. 125962.

[22] M. Ahmad et al., "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[23] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[24] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based k-nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, Nov. 2010, Art. no. 5555996.

[25] L. Shen and S. Jia, "Three-dimensional Gabor wavelets for pixel-based hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 5039–5046, Dec. 2011.

[26] M. D. Mura, J. A. Benediktsson, B. Waske, and L. Bruzzone, "Morphological attribute profiles for the analysis of very high resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3747–3762, Oct. 2010.

[27] Y. Tarabalka, J. A. Benediktsson, and J. Chanussot, "Spectral–spatial classification of hyperspectral imagery based on partitional clustering techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 8, pp. 2973–2987, Aug. 2009.

[28] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, Jul. 2015, Art. no. 258619.

[29] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[30] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[31] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[32] Z. Zhao, X. Xu, J. Li, S. Li, and A. Plaza, "Gabor-modulated grouped separable convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518817.

[33] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.

[34] X. Li, M. Ding, and A. Pižurica, "Spectral feature fusion networks with dual attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508614.

[35] X. Zhang, S. Shang, X. Tang, J. Feng, and L. Jiao, "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507714.

[36] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[37] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[38] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[39] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[40] J. An, L. Dai, M. Wang, and W. Zhang, "Multiscale low-rank and sparse attention-based transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[41] P. Zhu and J. Liu, "Channel-wise transformer with spectral–spatial gated self-attention for hyperspectral image classification," *Opt. Laser Technol.*, vol. 193, Jan. 2026, Art. no. 114182.

[42] R. Xu, X.-M. Dong, W. Li, J. Peng, W. Sun, and Y. Xu, "DBCTNet: Double branch convolution-transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509915.

[43] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[44] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[45] S. Xu, R. Liu, B. Jia, Z. Xie, and H. Gao, "DBSSFNet: Dual branch spatial–spectral-fusion-net for hyperspectral image classification," *Opt. Laser Technol.*, vol. 194, Feb. 2026, Art. no. 114473.

[46] Y. Zhang et al., "S2DBFT: Spectral–spatial dual-branch fusion transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5525517.

[47] Z. Zhang, T. Li, X. Tang, X. Hu, and Y. Peng, "CAEVT: Convolutional autoencoder meets lightweight vision transformer for hyperspectral image classification," *Sensors*, vol. 22, no. 10, p. 3902, May 2022.

[48] J. Zou, W. He, and H. Zhang, "PSFormer: Pyramid superpixel transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532816.

[49] W. Zhou, L. Bai, D. Xue, Y. Zhang, A. Gozho, and M. Hui, "MsGsFNet: A multiscale spatial and global spectral information fusion network for UAV-H^2 imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5533416.

[50] J. Chen, W. Li, and H. El-Askary, "Spectral–spatial transformer with multiscale convolutional attention for hyperspectral image classification," *IEEE J. Sel. Top. Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 24223–24241, 2025.

[51] H. Feng et al., "Flexible exploration: A hyperspectral image classification framework based on dynamic scale and graph optimal transport," *Inf. Fusion*, vol. 128, Apr. 2026, Art. no. 103962.

[52] W. Chen, Y. Zhang, Z. Xiao, J. Chu, and X. Wang, "Spectral–spatial self-supervised learning for few-shot hyperspectral image classification," 2025, *arXiv:2505.12482*.

[53] W. Liu, Y. Zhang, and Y. Dong, "Multifeature collaborative attention dynamic hypergraph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5522115.

[54] Y. Hu, J. Ge, and S. Tian, "STM-net: A multiscale spectral–spatial representation hybrid CNN-transformer model for hyperspectral image classification," *Remote Sens.*, vol. 17, no. 24, p. 4031, Dec. 2025.

[55] X. Dai, Z. Li, L. Li, S. Xue, X. Huang, and X. Yang, "HyperTransXNet: Learning both global and local dynamics with a dual dynamic token mixer for hyperspectral image classification," *Remote Sens.*, vol. 17, no. 14, p. 2361, Jul. 2025.

[56] N. Jiang, S. Geng, Y. Zheng, and L. Sun, "MSDCA: A multi-scale dual-branch network with enhanced cross-attention for hyperspectral image classification," *Remote Sens.*, vol. 17, no. 13, p. 2198, Jun. 2025.

[57] J. Liang and Y. Gao, "Lightweight memory-driven self-attention for hyperspectral image classification with CNN-transformer cross-feature fusion," *Neurocomputing*, vol. 651, Oct. 2025, Art. no. 130998.

[58] X. Liao, F. Lei, X. Liu, L. Guo, A. H.-M. Ng, and J. Ren, "An attention architecture with twice attention convolution and simplified transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[59] C. Guo, H. Huang, Z. Li, and T. Wang, "MSSL: Manifold geometry-leveraged self-supervised learning for hyperspectral image classification," *Electronics*, vol. 14, no. 24, p. 4935, Dec. 2025.

[60] P. Chen, W. He, F. Qian, G. Shi, and J. Yan, "A synergistic CNN-transformer network with pooling attention fusion for hyperspectral image classification," *Digit. Signal Process.*, vol. 160, May 2025, Art. no. 105070.

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